LECTURE: 2 BASIC OF C++ (PART - 3)

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What is Pointer?

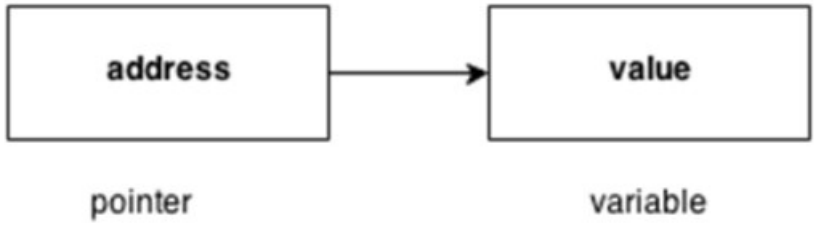
- A pointer is a variable that stores a memory address.
- Refers to some address in the program's memory space.
- Used to store the addresses of other variables or memory items.
- Very useful for another type of parameter passing, usually referred to as **Pass By Address**.
- Essential for dynamic memory allocation.

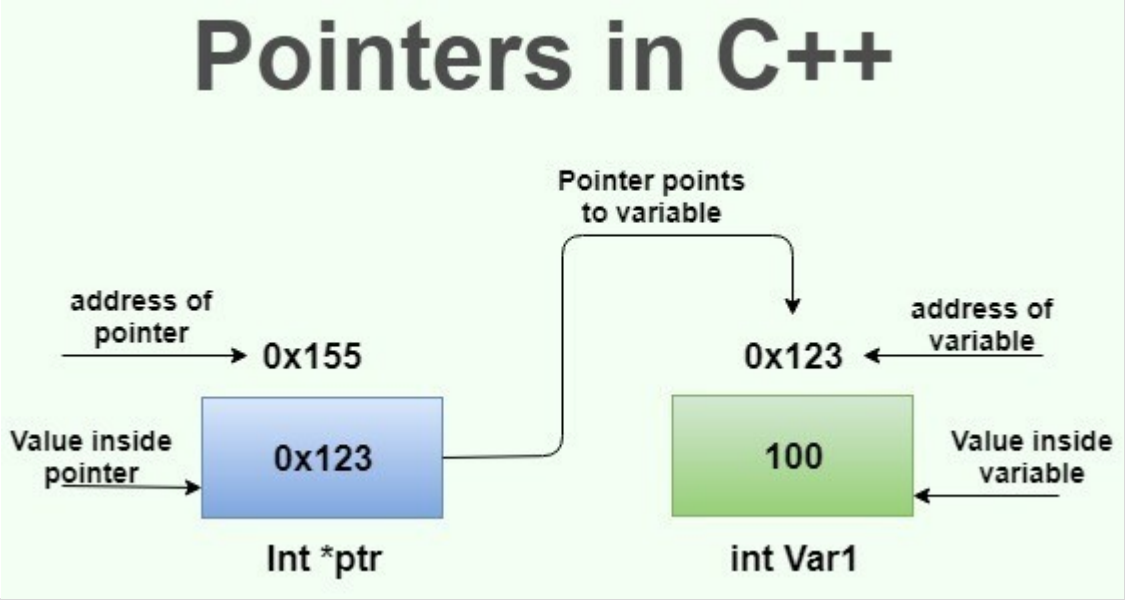
Why use Pointer?

- Short Ans: Don’t! ⏪ If it’s not necessary then don’t use it.
- Execution time with pointers is faster because data are manipulated with the address, that is, direct access to memory location.
- The pointer assigns and releases the memory as well. Hence it can be said the Memory of pointers is dynamically allocated.
- Pointers are used with data structures. They are useful for representing two-dimensional and multi-dimensional arrays.
- An array, of any type, can be accessed with the help of pointers, without considering its subscript range.
- Pointers are used for file handling.

Pointer Example 1

```
#include <iostream>
#include <string>
using namespace std;

int main() {
  int Var1 = 100;

  cout <<"Var1 = " <<Var1 << "\n";
  cout <<"&Var1 = " << &Var1 << "\n\n";

  int *ptr = &Var1;

  cout << "ptr = " << ptr << "\n";
  cout << "*ptr = " << *ptr << "\n";

  return 0;
}
```





Pointer Example 2

```
#include <iostream>
#include <string>
using namespace std;

int main() {
  string A = "NDUB Programming Club";  // A string variable
  string *ptr = &A;  // A pointer variable that stores the address of A

  // Output the value of A
  cout << A << "\n";

  // Output the memory address of A
  cout << &A << "\n";

  // Output the memory address of A with the pointer
  cout << ptr << "\n";

  // Output the data of A with the pointer
  cout << *ptr << "\n";
  return 0;
}
```

Notes

- Note that the * sign can be confusing here, as it does two different things in our code

- When used in declaration (string* ptr), it creates a **pointer variable**.

- When not used in declaration, it act as a **dereference operator**.

- The **dereferencing operator (*)** is a unary operator that returns the value present at the specified address.
- The **referencing operator (&)** is a unary operator that returns the address of its operand variable.

Swap two numbers using Pointers

```
#include <bits/stdc++.h>
using namespace std;
int main()
{
int a=20,b=10,*p1=&a,*p2=&b;

cout<<"Before swap: *p1="<<*p1<<" *p2="<<*p2<<endl;
*p1=*p1+*p2;
*p2=*p1-*p2;
*p1=*p1-*p2;


cout<<"After swap: *p1="<<*p1<<" *p2="<<*p2<<endl;
   return 0;
}
```

Null Pointer

- There is a special pointer whose value is 0. It is called the *null pointer*
- You **can** assign 0 into a pointer
- The null pointer is **never** a valid target, however. If you try to dereference the null pointer, you WILL get a segmentation fault
- So why use it? The null pointer is typically used as a placeholder to initialize pointers until you are ready to use them (i.e. with valid targets), so that their values are **known**

Null Pointer Example

```
#include <iostream>
using namespace std;

int main()
{
    int* ptr = NULL; //you can use both NULL and nullptr to create null pointer
    // Checking if the pointer is null before dereferencing
    if (ptr == nullptr) {
        cout << "Pointer is currently null." << endl;
    }
    else {
        cout << "Pointer is not null." << endl;
    }

    // *ptr = 10; (to avoid runtime error)
    // Assigning a valid memory address to the pointer
    int value = 5;
    ptr = &value;

    // Checking if the pointer is null after assigning a
    // valid address
    if (ptr == nullptr) {
        cout << "Pointer is currently null." << endl;
    }
    else {
        cout << "Pointer is not null." << endl;
        cout << "Value at the memory location pointed to "
                "by the pointer: " << *ptr << endl;
    }

    return 0;
}
```

Applications of Null Pointers

- **Initialization:** It is a good practice to Initialize pointers to a null value as it helps avoid undefined behavior by explicitly indicating they are not pointing to valid memory locations.

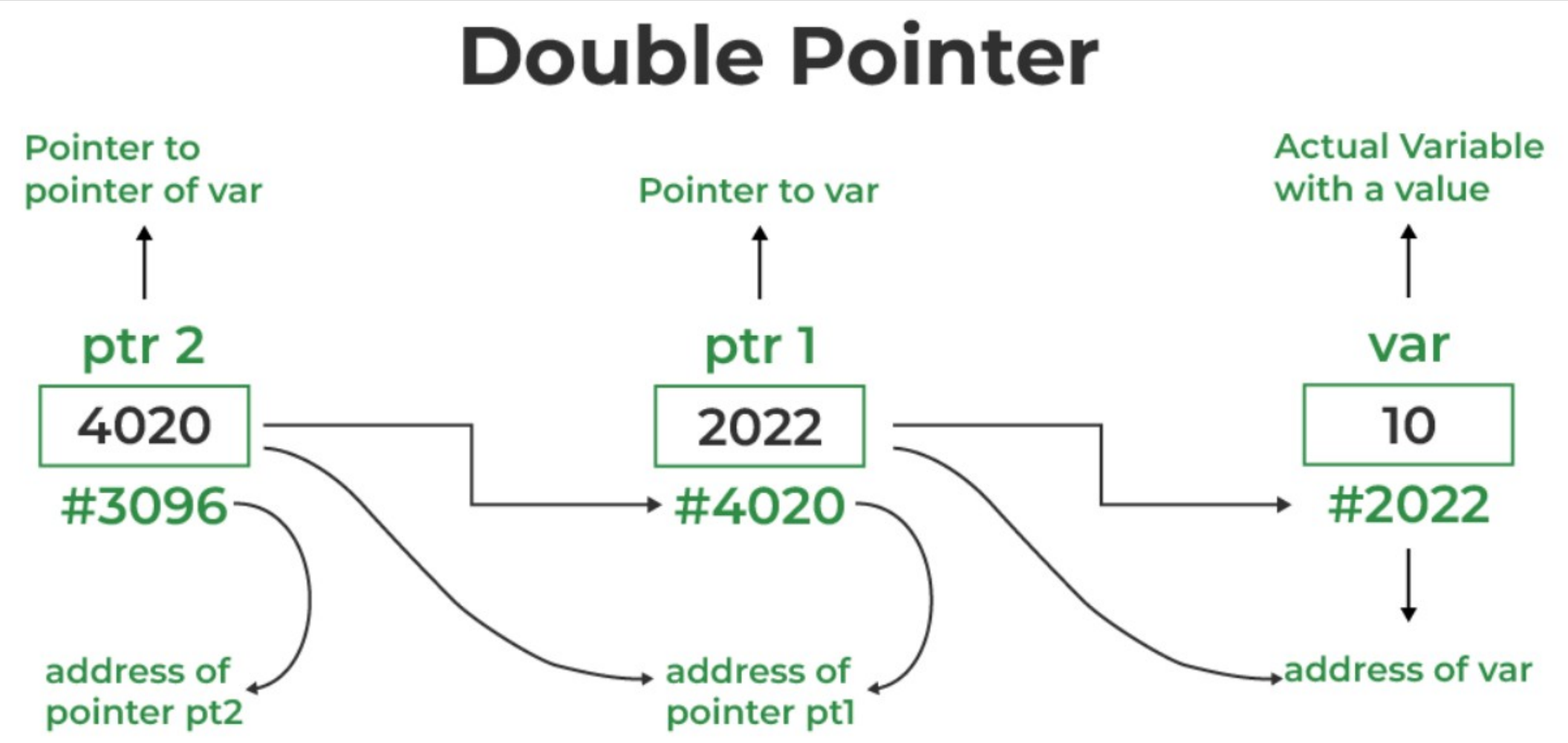
- **Default Values:** Null pointers act as default or initial values for pointers when no valid address is assigned to the pointers.

- **Error Handling:** They are useful in error conditions or to signify the absence of data that enables better handling of exceptional cases.

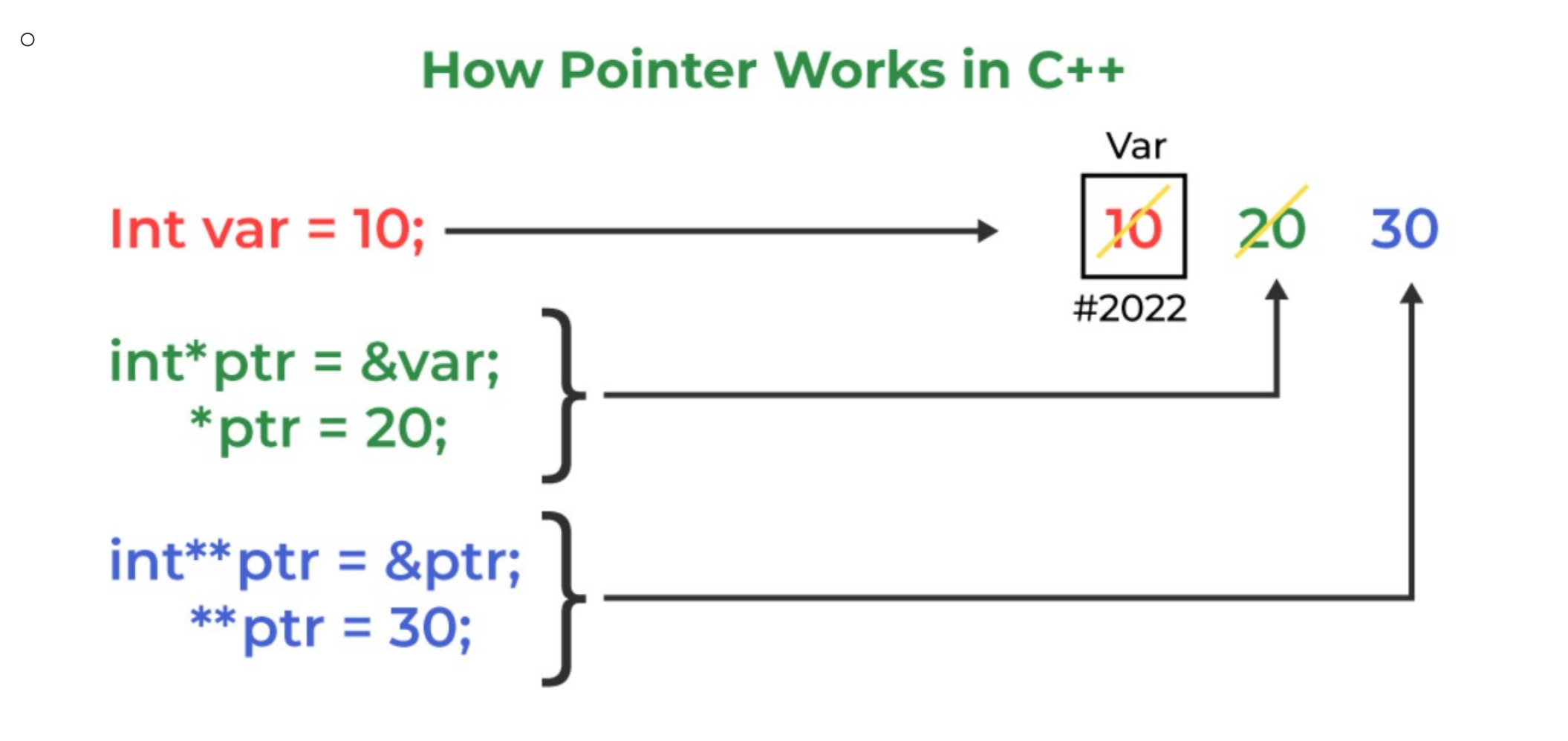
Double Pointer

- Now, we already know that a pointer stores the memory address of other variables.
- So, when we define a pointer to a pointer, the first pointer is used to store the address of the variables, and the second pointer stores the address of the first pointer.
- For this very reason, this is known as a **Double Pointer** or **Pointer to Pointer**

Double Pointer



Double Pointer



Double Pointer Example

```
#include <bits/stdc++.h>
using namespace std;

int main()
{
  int variable = 169;

  // Pointer to store the address
  // of variable
  int* pointer1;
  // double pointer to store the
  // address of pointer1
  int** pointer2;
  // Storing address of variable
  // in pointer1
  pointer1 = &variable;
  // Storing address of pointer1
  // in pointer2
  pointer2 = &pointer1;
  cout << "Value of variable :- " <<
          variable << "\n";
  cout << "Value of variable using single pointer :- " <<
          *pointer1 << "\n";
  cout << "Value of variable using double pointer :- " <<
          **pointer2 << "\n";
  return 0;
}
```

Array using Pointers

```
#include<bits/stdc++.h>

using namespace std;

int main()
{
  int arr[5] = { 1, 2, 3, 4, 5 };
  int *ptr = arr;

  cout << ptr << endl;
  // Get the value of the second element in ptr
  cout << *(ptr + 1) << endl;
  // Get the value of the third element in ptr
  cout << *(ptr + 2) << endl;

  //Using for loop to access array
  for (int i = 0; i < 5; i++)
  {
      cout << *(ptr + i) << endl;
  }
  return 0;
}
```

References:

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5. https://www.geeksforgeeks.org/cpp-pointer-to-pointer-double-pointer/
6. https://www.geeksforgeeks.org/what-is-dynamic-memory-allocation/